

Task 1: Shift Cipher Encryption

- Write a Python program that:
 - Takes a message (plaintext) from the user.
 - Asks for a numeric key (e.g., 5).
 - Encrypts the message by shifting **each alphabet letter forward** by that key.
 - Decrypts the resulting ciphertext back to the original message.
- Ignore non-alphabetic characters during shifting (e.g., spaces or punctuation remain unchanged).
- **Test your program** by encrypting the following:
“YourName is doing Security Lab ” with a **key = 5**.

Task 2: Random Letter Substitution Cipher

- Generate a random mapping between each alphabet letter (A–Z) and its substitute letter.
- Use this mapping to:
 - Encrypt a user-entered message.
 - Decrypt it back to the original message using the same key map.
- The mapping should change each time the program runs (demonstrate randomness).
- **Test your program** by encrypting and decrypting **your full name** “your full name”.

Task 3: Repeating-Shift Cipher (Vigenère Cipher)

Question for Students:

- Encrypt the name “your name” using the **keyword** “SECURE”.
- Fill in a table showing: Plaintext letters, repeated keyword letters, numeric shifts, and resulting ciphertext.
- Then decrypt the ciphertext to verify you get the original message.

Table to Fill (Students Use):

Plaintext (P)						
Keyword (K)						
Shift (0–25)						
Ciphertext (C)						

Solved Example Using “WASIFA” as hint:

Step 1: Repeat the keyword to match message length if keyword length is smaller than your name length:

- Keyword: SECURE

Step 2: Convert letters to numbers (A=0 ... Z=25)

Plaintext (P)	W	A	S	I	F	A
Keyword (K)	S	E	C	U	R	E
Shift (0–25)	18	4	2	20	17	4

Step 3: Encrypt: $C_i = (P_i + K_i) \bmod 26$

- $W(22) + S(18) = 14 \rightarrow O$
- ----- so on till last letter of name

| Ciphertext (C) | OEUCWE

Step 4: Decrypt: $P_i = (C_i - K_i) \bmod 26$

- $O(14) - S(18) = 22 \rightarrow W$
- Original message: **WASIFA**

